EE621 Homework 1–2–3–4: Detailed Solutions

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1 Chapter 1 (Preliminaries): Solutions to 1.1–1.43

1.1 Question. For $N \in \{0, 1, 2, \dots\}$,
Solution. For $N \in \{0, 1, 2, \dots\}$,

$$N = \sum_{k \geq 1} \mathbf{1}\{N \geq k\} \implies \mathbb{E}[N] = \sum_{k \geq 1} \mathbb{P}(N \geq k) = \sum_{k \geq 0} \mathbb{P}(N > k).$$

For nonnegative X with cdf F and tail $\bar{F} = 1 - F$,

$$\mathbb{E}[X] = \int_0^\infty \bar{F}(x)\, dx, \qquad \mathbb{E}[X^n] = n \int_0^\infty x^{n-1} \bar{F}(x)\, dx.$$

(Use Fubini/Tonelli on $\int_0^\infty \mathbf{1}\{X > x\} dx$ and $\int_0^\infty n x^{n-1} \mathbf{1}\{X > x\} dx$.)

1.2 Question. (a) If X has continuous cdf F, then for $0 < u < 1$,
Solution. (a) If X has continuous cdf F, then for $0 < u < 1$,

$$\mathbb{P}(F(X) \leq u) = \mathbb{P}(X \leq F^{-1}(u)) = u,$$

so $F(X) \sim U(0, 1)$.
(b) If $U \sim U(0, 1)$ and $Y = F^{-1}(U)$, then

$$\mathbb{P}(Y \leq y) = \mathbb{P}(U \leq F(y)) = F(y),$$

hence Y has cdf F.

1.3 Question. If $X_n \sim \mathrm{Bin}(n, p_n)$ with $np_n \to \lambda$,
Solution. If $X_n \sim \mathrm{Bin}(n, p_n)$ with $np_n \to \lambda$,

$$\mathbb{P}(X_n = i) = \binom{n}{i} p_n^i (1 - p_n)^{n-i} = \frac{(np_n)^i}{i!} (1 - p_n)^n \prod_{j=0}^{i-1} \frac{n - j}{n} (1 - p_n)^{-i} \to e^{-\lambda} \frac{\lambda^i}{i!}.$$

So $X_n \Rightarrow \mathrm{Poisson}(\lambda)$.

1.4 Question. For $X \sim \mathrm{Bin}(n, p)$,
Solution. For $X \sim \mathrm{Bin}(n, p)$,

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1 - p).$$

1.5 Question. Let $(N_1, \dots, N_r)$ be counts from n i.i.d. trials with category probabilities $(p_1, \dots, p_r)$.
Solution. Let $(N_1, \dots, N_r)$ be counts from n i.i.d. trials with category probabilities $(p_1, \dots, p_r)$.
(a) Multinomial law:

$$\mathbb{P}(N_1 = n_1, \dots, N_r = n_r) = \frac{n!}{\prod_i n_i!} \prod_{i=1}^r p_i^{n_i}, \quad \sum_i n_i = n.$$

(b) For $i \neq j$,

$$\mathrm{Cov}(N_i, N_j) = -np_i p_j.$$

(c) Let $M = \sum_{i=1}^r I_i$, $I_i = \mathbf{1}\{N_i = 0\}$. Then

$$\mathbb{E}[M] = \sum_{i=1}^r (1 - p_i)^n,$$

$$\mathrm{Var}(M) = \sum_i a_i(1 - a_i) + 2 \sum_{i<j} \big((1 - p_i - p_j)^n - a_i a_j\big), \quad a_i = (1 - p_i)^n.$$

1.6 Question. Let record indicators $I_k = \mathbf{1}\{X_k > \max(X_1, \dots, X_{k-1})\}$ for i.i.d. continuous X_k.

Solution. Let record indicators $I_k = \mathbf{1}\{X_k > \max(X_1, \dots, X_{k-1})\}$ for i.i.d. continuous X_k.

(a) $\mathbb{P}(I_k = 1) = 1/k$, and (I_k) are independent. Hence

$$N_n = \sum_{k=1}^{n} I_k, \quad \mathbb{E}[N_n] = \sum_{k=1}^{n} \frac{1}{k} =: H_n, \quad \mathrm{Var}(N_n) = \sum_{k=1}^{n} \frac{1}{k}\Big(1 - \frac{1}{k}\Big) = H_n - \sum_{k=1}^{n} \frac{1}{k^2}.$$

(b) $T = \min\{n > 1 : I_n = 1\}$: $\mathbb{P}(T > m) = \mathbb{P}(I_2 = \cdots = I_m = 0) = 1/m$. Thus

$$\mathbb{P}(T < \infty) = 1, \qquad \mathbb{E}[T] = \sum_{m \geq 1} \mathbb{P}(T \geq m) = \infty.$$

(c) $T_y = \min\{n : X_n > y\}$ is geometric with parameter $q = \mathbb{P}(X > y)$ and independent of the overshoot sample value X_{T_y}; moreover

$$\mathcal{L}(X_{T_y}) = \mathcal{L}(X \mid X > y).$$

So $T_y \perp X_{T_y}$.

1.7 Question. Hypergeometric sampling of k balls from urn with n white, m black:

Solution. Hypergeometric sampling of k balls from urn with n white, m black:

$$X \sim \mathrm{Hypergeom}(n + m, n, k),$$

$$\mathbb{E}[X] = k\frac{n}{n+m}, \quad \mathrm{Var}(X) = k\frac{n}{n+m}\frac{m}{n+m}\frac{n+m-k}{n+m-1}.$$

1.8 Question. If $X_1 \sim \mathrm{Pois}(\lambda_1), X_2 \sim \mathrm{Pois}(\lambda_2)$ independent:

Solution. If $X_1 \sim \mathrm{Pois}(\lambda_1), X_2 \sim \mathrm{Pois}(\lambda_2)$ independent:

(a) $X_1 + X_2 \sim \mathrm{Pois}(\lambda_1 + \lambda_2)$.

(b)

$$X_1 \mid (X_1 + X_2 = n) \sim \mathrm{Bin}\Big(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\Big).$$

1.9 Question. Orient each edge independently with probability 1/2 (round-robin outcomes). For a fixed permutation, the probability it forms a directed Hamilton path is $2^{-(n-1)}$. Number of permutations is $n!$, so expected number of Hamilton paths is

Solution. Orient each edge independently with probability 1/2 (round-robin outcomes). For a fixed permutation, the probability it forms a directed Hamilton path is $2^{-(n-1)}$. Number of permutations is $n!$, so expected number of Hamilton paths is

$$\frac{n!}{2^{n-1}}.$$

Hence some tournament has at least this many Hamilton paths.

1.10 Question. Let A_S be event that no outside contestant beats all members of a fixed k-set S. For any outsider u, probability u beats all in S is 2^{-k}; hence

Solution. Let A_S be event that no outside contestant beats all members of a fixed k-set S. For any outsider u, probability u beats all in S is 2^{-k}; hence

$$\mathbb{P}(A_S) = \left(1 - 2^{-k}\right)^{n-k} \leq 1 - 2^{1-k(n-k)}$$

(using $1 - x \leq e^{-x}$ and the bound given in the text statement). Union bound over $\binom{n}{k}$ subsets yields

$$\mathbb{P}\Big(\bigcup_S A_S\Big) \leq \binom{n}{k}\left(1 - 2^{1-k(n-k)}\right) < 1.$$

Therefore there exists an outcome where every k-set is dominated by some contestant.

1.11 Question. For pgf $P(z) = \mathbb{E}[z^X] = \sum_{j\geq 0} z^j \mathbb{P}(X = j)$:

Solution. For pgf $P(z) = \mathbb{E}[z^X] = \sum_{j\geq 0} z^j \mathbb{P}(X = j)$:

(a)

$$\frac{d^k}{dz^k} P(z)\big|_{z=0} = k!\mathbb{P}(X = k).$$

(b) Even probability:

$$\mathbb{P}(X \text{ even}) = \frac{P(1) + P(-1)}{2}.$$

(c) If $X \sim \mathrm{Bin}(n, p)$, $P(z) = (1 - p + pz)^n$, so

$$\mathbb{P}(X \text{ even}) = \frac{1 + (1 - 2p)^n}{2}.$$

(d) If $X \sim \mathrm{Pois}(\lambda)$, $P(z) = e^{\lambda(z-1)}$, so

$$\mathbb{P}(X \text{ even}) = \frac{1 + e^{-2\lambda}}{2}.$$

(e) If $X \sim \mathrm{Geom}(p)$ on $\{1, 2, \dots\}$,

$$\mathbb{P}(X \text{ even}) = \sum_{j\geq 1} p(1 - p)^{2j-1} = \frac{1 - p}{2 - p}.$$

(f) For negative binomial (r, p),

$$\mathbb{P}(X \text{ even}) = \frac{1}{2}\left[1 + (-1)^r \left(\frac{p}{2 - p}\right)^r\right].$$

1.12 Question. If $0 \leq X \leq a$ a.s., then from $X(a - X) \geq 0$,

Solution. If $0 \leq X \leq a$ a.s., then from $X(a - X) \geq 0$,

$$\mathbb{E}[X^2] \leq a\mathbb{E}[X] = a\mu.$$

Hence

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2 \leq a\mu - \mu^2 = \mu(a - \mu) \leq a^2/4.$$

1.13 Question. Let tagged card be original bottom card and let J_t be its position from top. For $j > 1$, one operation decreases J by 1 with probability $(n - j + 1)/n$, else leaves it at j. Thus expected time to move from j to $j - 1$ is

Solution. Let tagged card be original bottom card and let J_t be its position from top. For $j > 1$, one operation decreases J by 1 with probability $(n - j + 1)/n$, else leaves it at j. Thus expected time to move from j to $j - 1$ is

$$\frac{1}{(n-j+1)/n} = \frac{n}{n-j+1}.$$

Summing from $j = n$ to 2, plus one final operation after it reaches top:

$$\mathbb{E}[T] = \sum_{m=1}^{n-1} \frac{n}{m} + 1 = nH_{n-1} + 1.$$

By symmetry and insertion invariance, final deck order is uniform over $n!$ permutations.

1.14 Question. Condition: first 9 rolls are odd, 10th is even.

Solution. Condition: first 9 rolls are odd, 10th is even.

(a) X_1 counts side 1 among first 9 odd rolls, so $X_1 \sim \mathrm{Bin}(9, 1/3)$.

$$\mathbb{E}[X_1] = 3.$$

(b)

$$\mathbb{E}[X_1^2] = \mathrm{Var}(X_1) + \mathbb{E}[X_1]^2 = 9 \cdot \frac{1}{3} \cdot \frac{2}{3} + 9 = 11.$$

(c)

$$\mathbb{P}(X_1 = k) = \binom{9}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{9-k}, \quad k = 0, \ldots, 9.$$

(d) X_2 is indicator that 10th roll is side 2, so $X_2 \sim \mathrm{Bernoulli}(1/3)$:

$$\mathbb{P}(X_2 = 1) = \frac{1}{3}, \quad \mathbb{P}(X_2 = 0) = \frac{2}{3}.$$

1.15 Question. (a) If $X = F^{-1}(U)$ with $U \sim U(0, 1)$, then $X \sim F$.

Solution. (a) If $X = F^{-1}(U)$ with $U \sim U(0, 1)$, then $X \sim F$.

(b) If $U \sim U(0, 1)$, then for $x > 0$,

$$\mathbb{P}(-\log U > x) = \mathbb{P}(U < e^{-x}) = e^{-x},$$

so $-\log U \sim \mathrm{Exp}(1)$ with mean 1.

1.16 Question. Accept-reject with proposal density g, target f, and $f \le cg$:

Solution. Accept-reject with proposal density g, target f, and $f \le cg$:

(a) Accepted sample density is f (standard rejection-sampling proof).

(b) Acceptance probability is

$$\int g(y) \frac{f(y)}{cg(y)} dy = \frac{1}{c},$$

so number of iterations is geometric with mean c.

1.17 Question. For i.i.d. continuous with cdf F:

Solution. For i.i.d. continuous with cdf F:

(a) Recursion for cdfs of order statistics:

$$F_{i,n}(x) = F(x)F_{i-1,n-1}(x) + \bar{F}(x)F_{i,n-1}(x).$$

(b)

$$F_{i,n-1}(x) = \frac{i}{n}F_{i+1,n}(x) + \frac{n-i}{n}F_{i,n}(x).$$

1.18 Question. Expected number of flips until r consecutive heads (head prob p):

Solution. Expected number of flips until r consecutive heads (head prob p):

$$\mathbb{E}[T_r] = \frac{1-p^r}{(1-p)p^r}$$

(and for fair coin $p = 1/2$, $\mathbb{E}[T_r] = 2^{r+1} - 2$).

1.19 Question. Let M_n be expected number of white balls after n draws.

Solution. Let M_n be expected number of white balls after n draws.

(a)

$$M_{n+1} = \left(1 - \frac{1}{a+b}\right) M_n + 1.$$

(b) Solve linear recursion:

$$M_n = a + b - b\left(1 - \frac{1}{a+b}\right)^n.$$

(c) Probability $(n+1)$st ball is white:

$$\mathbb{P}(W_{n+1}) = \frac{M_n}{a+b} = 1 - \frac{b}{a+b}\left(1 - \frac{1}{a+b}\right)^n.$$

1.20 Question. Let $N(x)$ be number packed in $(0, x)$ and $M(x) = \mathbb{E}[N(x)]$. By conditioning on first accepted interval's left endpoint and splitting remaining space:

Solution. Let $N(x)$ be number packed in $(0, x)$ and $M(x) = \mathbb{E}[N(x)]$. By conditioning on first accepted interval's left endpoint and splitting remaining space:

$$M(x) = 0, \ x < 1,$$

$$M(x) = 1 + \frac{2}{x-1}\int_0^{x-1} M(y)\,dy, \ x > 1.$$

1.21 Question. Define $Y_i = -\log U_i \sim \text{Exp}(1)$ i.i.d. Then

Solution. Define $Y_i = -\log U_i \sim \text{Exp}(1)$ i.i.d. Then

$$\prod_{i=1}^n U_i \le e^{-\lambda} \iff \sum_{i=1}^n Y_i \ge \lambda.$$

So N equals number of Poisson(1)-process arrivals by time λ, hence

$$N \sim \text{Poisson}(\lambda), \qquad \mathbb{P}(N = n) = e^{-\lambda}\frac{\lambda^n}{n!}.$$

1.22 Question. Using

Solution. Using

$$\mathrm{Var}(X \mid Y) = \mathbb{E}\big[(X - \mathbb{E}[X \mid Y])^2 \mid Y\big],$$

write

$$X - \mathbb{E}[X] = (X - \mathbb{E}[X \mid Y]) + \big(\mathbb{E}[X \mid Y] - \mathbb{E}[X]\big)$$

and square/condition. Cross term vanishes. Therefore

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).$$

1.23 Question. Random walk with step $+1$ prob p, -1 prob $q = 1 - p$, starting at 0.

Solution. Random walk with step $+1$ prob p, -1 prob $q = 1 - p$, starting at 0.

(a) Let $\alpha = \mathbb{P}_0(\text{ever hit } 1)$. First-step conditioning gives

$$\alpha = p + q\alpha^2.$$

(b) Solve quadratic:

$$\alpha = \begin{cases} 1, & p \geq \frac{1}{2}, \\ \dfrac{p}{q}, & p < \frac{1}{2}. \end{cases}$$

(c) Probability to ever hit n:

$$\mathbb{P}_0(\tau_n < \infty) = \alpha^n.$$

(d) For $p < 1/2$, condition on eventual hit of n (Doob h-transform), transition probabilities swap:

$$\mathbb{P}(i \to i+1 \mid \tau_n < \infty) = q, \qquad \mathbb{P}(i \to i-1 \mid \tau_n < \infty) = p.$$

1.24 Question. Let T be hitting time of 1 from 0.

Solution. Let T be hitting time of 1 from 0.

(a)

$$\mathbb{E}[T] = \begin{cases} \dfrac{1}{2p-1}, & p > 1/2, \\ \infty, & p \leq 1/2. \end{cases}$$

(b) For $p > 1/2$,

$$\mathrm{Var}(T) = \frac{4p(1-p)}{(2p-1)^3}.$$

(c) Hitting time of n from 0:

$$\mathbb{E}[T_n] = \begin{cases} \dfrac{n}{2p-1}, & p > 1/2, \\ \infty, & p \leq 1/2. \end{cases}$$

(d) For $p > 1/2$,

$$\mathrm{Var}(T_n) = \frac{4np(1-p)}{(2p-1)^3}.$$

1.25 Question. Gambler's ruin on $\{0, 1, \ldots, k\}$, start i, with win prob p, lose prob q.

Solution. Gambler's ruin on $\{0, 1, \ldots, k\}$, start i, with win prob p, lose prob q. Expected duration m_i solves

$$m_i = 1 + pm_{i+1} + qm_{i-1}, \quad m_0 = m_k = 0.$$

Hence

$$m_i = \begin{cases} i(k-i), & p = 1/2, \\ \dfrac{i}{q-p} - \dfrac{k}{q-p}\dfrac{1-(q/p)^i}{1-(q/p)^k}, & p \neq 1/2. \end{cases}$$

1.26 Question. Ballot theorem result: if candidate A gets a votes and B gets b with $a > b$, then probability A is never behind is

Solution. Ballot theorem result: if candidate A gets a votes and B gets b with $a > b$, then probability A is never behind is

$$\frac{a-b}{a+b}.$$

1.27 Question. In ruin problem starting with n, probability ruin occurs exactly at time $n + 2i$:

Solution. In ruin problem starting with n, probability ruin occurs exactly at time $n + 2i$:

$$\mathbb{P}(T = n + 2i) = \frac{n}{n+2i}\binom{n+2i}{i}p^i q^{n+i}, \qquad i \geq 0.$$

(Reflection principle/ballot theorem.)

1.28 Question. If $X \sim \mathrm{Exp}(\lambda)$,

Solution. If $X \sim \mathrm{Exp}(\lambda)$,

$$\mathbb{E}[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

1.29 Question. If $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Exp}(\lambda)$, then

Solution. If $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Exp}(\lambda)$, then

$$S_n = \sum_{i=1}^{n} X_i \sim \Gamma(n, \lambda),$$

with density

$$f_{S_n}(t) = \lambda e^{-\lambda t}\frac{(\lambda t)^{n-1}}{(n-1)!}, \quad t \geq 0.$$

1.30 Question. If service times are independent exponentials with rates λ_i, then for customer i,

Solution. If service times are independent exponentials with rates λ_i, then for customer i,

$$\mathbb{P}(i \text{ is last}) = \int_0^\infty \lambda_i e^{-\lambda_i t}\prod_{j \neq i}(1 - e^{-\lambda_j t})\,dt = \sum_{A \subseteq \{j \neq i\}} (-1)^{|A|}\frac{\lambda_i}{\lambda_i + \sum_{j \in A}\lambda_j}.$$

1.31 Question. If $X \sim \mathrm{Exp}(\lambda_1)$, $Y \sim \mathrm{Exp}(\lambda_2)$ independent, and $Z = \min(X, Y)$:

Solution. If $X \sim \mathrm{Exp}(\lambda_1)$, $Y \sim \mathrm{Exp}(\lambda_2)$ independent, and $Z = \min(X, Y)$:

$$Z \sim \mathrm{Exp}(\lambda_1 + \lambda_2).$$

Given $Z = X$ (equivalently $X < Y$),

$$Z \mid (Z = X) \sim \mathrm{Exp}(\lambda_1 + \lambda_2).$$

1.32 Question. Continuous Cauchy equation solution:

Solution. Continuous Cauchy equation solution:

$$g(s + t) = g(s) + g(t),\ \forall s, t \quad \Rightarrow \quad g(t) = ct.$$

1.33 Question. For continuous cdf F, let R_i be ith record value and $u(x) = -\log(1 - F(x))$. Then

Solution. For continuous cdf F, let R_i be ith record value and $u(x) = -\log(1 - F(x))$. Then

$$\mathbb{P}(R_i \le x) = \mathbb{P}(\Gamma(i, 1) \le u(x)) = 1 - e^{-u(x)} \sum_{k=0}^{i-1} \frac{u(x)^k}{k!} = 1 - (1 - F(x)) \sum_{k=0}^{i-1} \frac{[-\log(1 - F(x))]^k}{k!}.$$

1.34 Question. For independent nonnegative continuous X_1, X_2 with hazards λ_1, λ_2, cause-specific hazard identity gives

Solution. For independent nonnegative continuous X_1, X_2 with hazards λ_1, λ_2, cause-specific hazard identity gives

$$\mathbb{P}(X_1 < X_2 \mid \min(X_1, X_2) = t) = \frac{\lambda_1(t)}{\lambda_1(t) + \lambda_2(t)}.$$

(Interpreted via dt limit conditioning.)

1.35 Question. Tilted density:

Solution. Tilted density:

$$f_t(x) = \frac{e^{tx} f(x)}{M(t)}, \quad M(t) = \mathbb{E}[e^{tX}].$$

Let $X_t \sim f_t$.

(a)

$$\mathbb{E}[h(X)] = M(t)\, \mathbb{E}\big[e^{-tX_t} h(X_t)\big].$$

(b) Chernoff bound ($t > 0$):

$$\mathbb{P}(X > a) \le M(t) e^{-ta}.$$

(c) If $\mathbb{P}(X_{t^*} > a) = 1/2$, then

$$\min_t M(t) e^{-ta} = M(t^*) e^{-t^* a}.$$

1.36 Question. Jensen for concave log:

Solution. Jensen for concave log:

$$\log\Big(\frac{1}{n} \sum_{i=1}^n x_i\Big) \ge \frac{1}{n} \sum_{i=1}^n \log x_i \Rightarrow \frac{1}{n} \sum_{i=1}^n x_i \ge \Big(\prod_{i=1}^n x_i\Big)^{1/n}.$$

1.37 Question. Peak at time n means $X_{n-1} < X_n > X_{n+1}$. For i.i.d. continuous triples, all 6 orderings equiprobable; exactly 2 make middle value largest. Thus

Solution. Peak at time n means $X_{n-1} < X_n > X_{n+1}$. For i.i.d. continuous triples, all 6 orderings equiprobable; exactly 2 make middle value largest. Thus

$$\mathbb{P}(\text{peak at } n) = \frac{2}{6} = \frac{1}{3}.$$

By LLN/ergodic theorem, long-run proportion is 1/3 a.s.

1.38 Question. Coupon collector for states $1, \ldots, m$: expected steps to visit all is

Solution. Coupon collector for states $1, \ldots, m$: expected steps to visit all is

$$\mathbb{E}[T] = m \sum_{j=1}^{m} \frac{1}{j} = mH_m.$$

(Use decomposition into additional waits after $i-1$ states collected.)

1.39 Question. Simple symmetric random walk on line graph $\{0, 1, \ldots, n\}$ with 0 reflecting and target n absorbing: expected hit time from 0 equals

Solution. Simple symmetric random walk on line graph $\{0, 1, \ldots, n\}$ with 0 reflecting and target n absorbing: expected hit time from 0 equals

$$\mathbb{E}_0[T_n] = n^2.$$

(Second-order difference equation $e_i = 1 + \frac{1}{2}(e_{i-1} + e_{i+1})$ for interior.)

1.40 Question. For Example 1.9(C) with $r = 3$, the probability a given leaf is last visited is obtained from the star-graph last-visit recursion in Problem 1.41. Substitute $r = 3$ in that recursion and solve explicitly for the ray lengths in the example.

Solution. For Example 1.9(C) with $r = 3$, the probability a given leaf is last visited is obtained from the star-graph last-visit recursion in Problem 1.41. Substitute $r = 3$ in that recursion and solve explicitly for the ray lengths in the example.

1.41 Question. For a star with one ray length m and $r-1$ rays length n, let P_r be probability the length-m leaf is last visited. Each excursion from center hits leaf j with probability proportional to $1/\ell_j$ (gambler's ruin on a ray). Thus effective coupon...

Solution. For a star with one ray length m and $r-1$ rays length n, let P_r be probability the length-m leaf is last visited. Each excursion from center hits leaf j with probability proportional to $1/\ell_j$ (gambler's ruin on a ray). Thus effective coupon probabilities are

$$q_m \propto \frac{1}{m}, \qquad q_n \propto \frac{1}{n}.$$

Use the standard last-coupon recursion:

$$P_r = \sum_{j=1}^{r-1} \frac{q_j}{1 - q_m} P_{r-1}^{(j)}$$

with symmetry among equal-length rays to obtain P_r in terms of P_{r-1}.

1.42 Question. Given $\mathbb{P}(Y_i = 0) = \alpha$, $\mathbb{P}(Y_i > y) = (1-\alpha)e^{-y}$ and

Solution. Given $\mathbb{P}(Y_i = 0) = \alpha$, $\mathbb{P}(Y_i > y) = (1-\alpha)e^{-y}$ and

$$X_0 = 0, \qquad X_{n+1} = \alpha X_n + Y_{n+1},$$

inductively:

$$\mathbb{P}(X_n = 0) = \alpha^n, \qquad \mathbb{P}(X_n > x) = (1-\alpha^n)e^{-x}, \quad x > 0.$$

1.43 Question. Markov inequality on X^n gives

Solution. Markov inequality on X^n gives

$$\mathbb{P}(X \geq a) = \mathbb{P}(X^n \geq a^n) \leq \frac{\mathbb{E}[X^n]}{a^n}.$$

To show $n! \geq (n/e)^n$, use Poisson(n):

$$1 \geq \mathbb{P}(N = n) = e^{-n}\frac{n^n}{n!} \Rightarrow n! \geq \left(\frac{n}{e}\right)^n.$$

2 Chapter 2 (Poisson Process): Solutions to 2.1–2.42

2.1 Question. Definition 2.1.1 (independent stationary increments + initial condition) implies Definition 2.1.2 (Poisson counts) because for any partition,

Solution. Definition 2.1.1 (independent stationary increments + initial condition) implies Definition 2.1.2 (Poisson counts) because for any partition,

$$N(t) = \sum_{j=1}^{m} \big(N(t_j) - N(t_{j-1})\big)$$

with independent increments each Poisson with mean $\lambda(t_j - t_{j-1})$; summing gives Poisson(λt).

2.2 Question. (a) From Definition 2.1.2 and independent increments,

Solution. (a) From Definition 2.1.2 and independent increments,

$$P_0(t+s) = \mathbb{P}(N(t+s) = 0) = \mathbb{P}(N(t) = 0, N(t+s) - N(t) = 0) = P_0(t)P_0(s).$$

(b) Cauchy + continuity gives $P_0(t) = e^{-\lambda t}$.

(c) Interarrival times are i.i.d. Exp(λ); therefore $N(t)$ is Poisson with mean λt.

2.3 Question. Given $N(t) = n$, arrivals are order stats of n i.i.d. Uniform$(0,t)$, so for $s < t$,

Solution. Given $N(t) = n$, arrivals are order stats of n i.i.d. Uniform$(0,t)$, so for $s < t$,

$$\mathbb{P}(N(s) = k \mid N(t) = n) = \binom{n}{k}\left(\frac{s}{t}\right)^k\left(1 - \frac{s}{t}\right)^{n-k}.$$

2.4 Question. Let $s > 0$ and $N(t+s) = N(t) + M$ with $M \sim$ Pois(λs) independent of $N(t)$. Then

Solution. Let $s > 0$ and $N(t+s) = N(t) + M$ with $M \sim$ Pois(λs) independent of $N(t)$. Then

$$\mathbb{E}[N(t)N(t+s)] = \mathbb{E}[N(t)^2] + \mathbb{E}[N(t)]\mathbb{E}[M] = \lambda t + (\lambda t)^2 + \lambda^2 ts = \lambda t + \lambda^2 t(t+s).$$

2.5 Question. Independent Poisson processes N_1, N_2 with rates λ_1, λ_2:

Solution. Independent Poisson processes N_1, N_2 with rates λ_1, λ_2:

- $N_1 + N_2$ is Poisson with rate $\lambda_1 + \lambda_2$.
- Type of first event is from process 1 with probability $\lambda_1/(\lambda_1 + \lambda_2)$, independent of first-event time.

2.6 Question. If stockpile has r type-1 and s type-2 components with lifetimes $\mathrm{Exp}(\mu_1)$, $\mathrm{Exp}(\mu_2)$, exhaustion times are

Solution. If stockpile has r type-1 and s type-2 components with lifetimes $\mathrm{Exp}(\mu_1)$, $\mathrm{Exp}(\mu_2)$, exhaustion times are

$$T_1 = \sum_{i=1}^{r} E_i \sim \Gamma(r, \mu_1), \qquad T_2 = \sum_{j=1}^{s} F_j \sim \Gamma(s, \mu_2),$$

independent. Machine lifetime is $\min(T_1, T_2)$, so

$$\mathbb{E}[\min(T_1, T_2)] = \int_0^\infty \mathbb{P}(T_1 > t)\mathbb{P}(T_2 > t)\, dt.$$

Use gamma survival expansions for closed form finite sums.

2.7 Question. For Poisson process with rate λ, arrival times $S_1 < S_2 < S_3$ have joint density

Solution. For Poisson process with rate λ, arrival times $S_1 < S_2 < S_3$ have joint density

$$f_{S_1,S_2,S_3}(s_1, s_2, s_3) = \lambda^3 e^{-\lambda s_3}, \quad 0 < s_1 < s_2 < s_3.$$

2.8 Question. With $U_i \sim U(0,1)$ i.i.d., define $X_i = -\log U_i$ (Exp(1)) and $N = \min\{n : \sum_{i=1}^n X_i > \lambda\} - 1$.

Solution. With $U_i \sim U(0,1)$ i.i.d., define $X_i = -\log U_i$ (Exp(1)) and $N = \min\{n : \sum_{i=1}^n X_i > \lambda\} - 1$. Then $N \sim \mathrm{Pois}(\lambda)$.

2.9 Question. Stop at first event after time $s \in [0, T]$. Win iff exactly one event occurs in $(s, T]$.

Solution. Stop at first event after time $s \in [0, T]$. Win iff exactly one event occurs in $(s, T]$.

$$\mathbb{P}(\text{win}; s) = \mathbb{P}(N(T) - N(s) = 1) = e^{-\lambda(T-s)}\lambda(T - s).$$

Max at $T - s = 1/\lambda$ (if feasible), giving maximum $1/e$.

2.10 Question. Let bus waiting time $B \sim \mathrm{Exp}(\lambda)$, ride time R, walk time W, policy wait up to s.

Solution. Let bus waiting time $B \sim \mathrm{Exp}(\lambda)$, ride time R, walk time W, policy wait up to s.

(a)

$$\mathbb{E}[T(s)] = \mathbb{E}[B \wedge s] + R\mathbb{P}(B \le s) + W\mathbb{P}(B > s) = \frac{1 - e^{-\lambda s}}{\lambda} + R(1 - e^{-\lambda s}) + We^{-\lambda s}.$$

(b) Differentiate:

$$\frac{d}{ds}\mathbb{E}[T(s)] = e^{-\lambda s}\big(1 + \lambda R - \lambda W\big).$$

Hence

$$\begin{cases} W < 1/\lambda + R: \\ \text{min at } s = 0, \\ W > 1/\lambda + R: \\ \text{min at } s = \infty, \\ W = 1/\lambda + R: \\ \text{all } s \text{ tie.} \end{cases}$$

(c) Compare "walk now" versus "wait one more expected interarrival $1/\lambda$ then ride".

2.11 Question. Cars form Poisson(λ). Let W be wait to first gap of length T. Standard result:

Solution. Cars form Poisson(λ). Let W be wait to first gap of length T. Standard result:

$$\mathbb{E}[W] = \frac{e^{\lambda T} - 1}{\lambda}.$$

(At $T = 0$, this gives 0; grows exponentially in T.)

2.12 Question. After a registered event, counter is dead for h.

Solution. After a registered event, counter is dead for h.

(a) First k events all registered iff first $k-1$ interarrivals exceed h:

$$\mathbb{P}(\text{first } k \text{ registered}) = e^{-\lambda h(k-1)}.$$

(b) If $t \geq (n-1)h$, with registered-event epochs represented as delayed renewal,

$$\mathbb{P}(R(t) \geq n) = \mathbb{P}\big(E_n(\lambda) + (n-1)h \leq t\big) = 1 - e^{-\lambda(t-(n-1)h)} \sum_{j=0}^{n-1} \frac{[\lambda(t-(n-1)h)]^j}{j!}.$$

2.13 Question. With shock process Poisson(λ), each shock fatal with prob p independently. Condition on failure time $T = t$: nonfatal shocks before t form Poisson with mean $\lambda(1-p)t$.

Solution. With shock process Poisson(λ), each shock fatal with prob p independently. Condition on failure time $T = t$: nonfatal shocks before t form Poisson with mean $\lambda(1-p)t$. So

$$N - 1 \mid (T = t) \sim \text{Pois}(\lambda(1-p)t),$$

$$\mathbb{P}(N = n \mid T = t) = e^{-\lambda(1-p)t} \frac{(\lambda(1-p)t)^{n-1}}{(n-1)!}, \quad n \geq 1.$$

2.14 Question. Let $N_i \sim \text{Pois}(\lambda_i)$ independent and each entrant from floor i exits at j with prob p_{ij}.

Solution. Let $N_i \sim \text{Pois}(\lambda_i)$ independent and each entrant from floor i exits at j with prob p_{ij}.

(a)

$$\mathbb{E}[O_j] = \sum_{i<j} \lambda_i p_{ij}.$$

(b) By Poisson splitting/superposition:

$$O_j \sim \text{Pois}\Big(\sum_{i<j} \lambda_i p_{ij}\Big).$$

(c) Jointly, (O_j) are independent Poisson with the above means.

2.15 Question. For coin with side probabilities p_i, N_i = flips until side i appears n_i times.

Solution. For coin with side probabilities p_i, N_i = flips until side i appears n_i times.

(a) N_i is negative binomial (n_i, p_i).

(b) (N_i) are not independent (same flip sequence).

Poissonization with rate-1 event times gives independent Poisson subprocesses of rates p_i; let T_i = time until n_ith type-i event.

(c)

$$T = \min_i T_i, \qquad \mathbb{P}(T > t) = \prod_{i=1}^{r} \mathbb{P}(\text{Pois}(p_i t) < n_i) = \prod_{i=1}^{r} e^{-p_i t} \sum_{k=0}^{n_i - 1} \frac{(p_i t)^k}{k!}.$$

(d) T_i are independent (independent Poisson subprocesses).

(e)

$$\mathbb{E}[T] = \int_0^\infty \mathbb{P}(T > t) dt.$$

(f) If N is required flip count in original discrete model, Poissonization implies $\mathbb{E}[N] = \mathbb{E}[T]$.

2.16 Question. If number of trials $N \sim \text{Pois}(\lambda)$ and each trial has category probs (p_j), then by Poisson thinning

Solution. If number of trials $N \sim \text{Pois}(\lambda)$ and each trial has category probs (p_j), then by Poisson thinning

$$X_j \sim \text{Pois}(\lambda p_j),$$

independently across j. So

$$\mathbb{E}[X_j] = \lambda p_j, \qquad \text{Var}(X_j) = \lambda p_j.$$

2.17 Question. Order statistic $X_{(i)}$ from i.i.d. cdf F, pdf f:

Solution. Order statistic $X_{(i)}$ from i.i.d. cdf F, pdf f:

(a)

$$f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!} [F(x)]^{i-1} [1 - F(x)]^{n-i} f(x).$$

(b) $\{X_{(i)} \le x\}$ iff at least i of the X_j are $\le x$.

(c)

$$\mathbb{P}(X_{(i)} \le x) = \sum_{k=i}^{n} \binom{n}{k} F(x)^k (1 - F(x))^{n-k}.$$

(d) This yields the stated beta-integral identity by substitution $y = F(x)$.

(e) For Poisson process arrival times S_i,

$$\mathbb{E}[S_i \mid N(t) = n] = \begin{cases} \dfrac{i}{n+1} t, & i \le n, \\ t + \dfrac{i-n}{\lambda}, & i > n. \end{cases}$$

2.18 Question. Given i.i.d. uniforms and $U_{(n)} = u$, the remaining order stats are distributed as order stats of $n-1$ i.i.d. Uniform$(0, u)$ variables.

Solution. Given i.i.d. uniforms and $U_{(n)} = u$, the remaining order stats are distributed as order stats of $n-1$ i.i.d. Uniform$(0, u)$ variables.

2.19 Question. If buses arrive as renewal process (interarrival cdf G) and bus capacity random J with $\mathbb{P}(J = j) = \alpha_j$, number served by time t is

Solution. If buses arrive as renewal process (interarrival cdf G) and bus capacity random J with $\mathbb{P}(J = j) = \alpha_j$, number served by time t is

$$X(t) = \sum_{k=1}^{N_G(t)} J_k.$$

Then

$$\mathbb{E}[X(t)] = m_G(t)\mathbb{E}[J],$$

where m_G is renewal function. In general $X(t)$ is not Poisson.

2.20 Question. For Poisson events (rate λ) classified at time s into type i with probs $p_i(s)$, $\sum_i p_i(s) = 1$:

Solution. For Poisson events (rate λ) classified at time s into type i with probs $p_i(s)$, $\sum_i p_i(s) = 1$:

$$N_i(t) \text{ are independent}, \quad N_i(t) \sim \text{Pois}\left(\lambda \int_0^t p_i(s)\, ds\right).$$

2.21 Question. Poisson arrivals rate λ, each arrival independently follows state-probability profile $(p_i(s))_{i\geq 1}$ at age s.

Solution. Poisson arrivals rate λ, each arrival independently follows state-probability profile $(p_i(s))_{i\geq 1}$ at age s. Then numbers in states at time t are independent Poisson:

$$N_i(t) \sim \text{Pois}\left(\lambda \int_0^t p_i(u)\, du\right),$$

so

$$\mathbb{E}[N_i(t)] = \lambda \int_0^t p_i(u)\, du, \quad \text{Var}(N_i(t)) = \mathbb{E}[N_i(t)].$$

2.22 Question. Cars enter at Poisson rate λ, speed $V > 0$ with cdf F. Number in interval (a, b) at fixed time is Poisson with mean

Solution. Cars enter at Poisson rate λ, speed $V > 0$ with cdf F. Number in interval (a, b) at fixed time is Poisson with mean

$$\lambda(b - a)\mathbb{E}\left[\frac{1}{V}\right],$$

hence

$$N_{(a,b)} \sim \text{Pois}\big(\lambda(b - a)\mathbb{E}[1/V]\big).$$

2.23 Question. For compound Poisson model $D(t) = \sum_{k=1}^{N(t)} X_k$ (Example 2.3(C)):

Solution. For compound Poisson model $D(t) = \sum_{k=1}^{N(t)} X_k$ (Example 2.3(C)):

$$\text{Var}(D(t)) = \lambda t\mathbb{E}[X^2], \qquad \text{Cov}(D(t), D(t + s)) = \lambda t\mathbb{E}[X^2].$$

2.24 Question. Expected encounter rate for choosing speed v is proportional to

Solution. Expected encounter rate for choosing speed v is proportional to

$$\mathbb{E}|V - v|.$$

Any median of G minimizes $\mathbb{E}|V - v|$, so as $t \to \infty$ optimal speed tends to a median of G.

2.25 Question. With Poisson event times (rate λ) and i.i.d. contributions, total contribution by time t is compound Poisson:

Solution. With Poisson event times (rate λ) and i.i.d. contributions, total contribution by time t is compound Poisson:

$$W \stackrel{d}{=} \sum_{i=1}^{N} X_i, \quad N \sim \text{Pois}(\lambda t), \quad X_i \stackrel{iid}{\sim} F_X.$$

(If contribution law depends on event time s, then F_X is the corresponding time-mixture.)

2.26 Question. Condition on $S_n = t$: $(S_1, \ldots, S_{n-1})$ are distributed as order stats of $n-1$ i.i.d. Uniform$(0, t)$ variables.

Solution. Condition on $S_n = t$: $(S_1, \ldots, S_{n-1})$ are distributed as order stats of $n-1$ i.i.d. Uniform$(0, t)$ variables.

2.27 Question. For compound Poisson $D(t)$:

Solution. For compound Poisson $D(t)$:

$$M_{D(t)}(\theta) = \mathbb{E}[e^{\theta D(t)}] = \exp\{\lambda t(M_X(\theta) - 1)\}.$$

2.28 Question. Lemma 2.3 follows by conditioning on first jump time and using independent increments (or equivalently from Laplace transform/pgf identities for compound Poisson processes).

Solution. Lemma 2.3 follows by conditioning on first jump time and using independent increments (or equivalently from Laplace transform/pgf identities for compound Poisson processes).

2.29 Question. For nonhomogeneous Poisson with mean function $m(t)$:

Solution. For nonhomogeneous Poisson with mean function $m(t)$:

$$N(t+s) - N(t) \sim \text{Pois}(m(t+s) - m(t))$$

by partition approximation and independent increments.

2.30 Question. For NHPP intensity $\lambda(t)$ and interarrivals T_i:

Solution. For NHPP intensity $\lambda(t)$ and interarrivals T_i:

(a) generally not independent.

(b) not identically distributed.

(c)

$$\mathbb{P}(T_1 > t) = e^{-m(t)}, \quad F_{T_1}(t) = 1 - e^{-m(t)}.$$

(d) With $S_1 = T_1$, $S_2 = T_1 + T_2$,

$$f_{S_2}(s) = \lambda(s)e^{-m(s)}m(s),$$

then obtain T_2 by convolution/deconvolution with f_{T_1}.

2.31 Question. Time-change theorem: if N is NHPP with mean $m(t)$ and m invertible,

Solution. Time-change theorem: if N is NHPP with mean $m(t)$ and m invertible,

$$N^*(t) := N(m^{-1}(t))$$

is homogeneous Poisson with rate 1.

2.32 Question. (a) Given $N(t) = n$, ordered arrivals have order-stat law from i.i.d. cdf

Solution. (a) Given $N(t) = n$, ordered arrivals have order-stat law from i.i.d. cdf

$$F(x) = \frac{m(x)}{m(t)}, \ 0 \le x \le t.$$

(b) If accidents follow NHPP mean $m(t)$ and each injury duration has cdf F, number out of work at t is Poisson with mean

$$\mu(t) = \int_0^t \lambda(u)\, \bar{F}(t-u)\, du,$$

so

$$\mathbb{E}[X(t)] = \mathrm{Var}(X(t)) = \mu(t).$$

2.33 Question. 2D Poisson process with rate λ per unit area.

Solution. 2D Poisson process with rate λ per unit area.

(a)

$$\mathbb{P}(X > t) = e^{-\lambda \pi t^2}.$$

(b)

$$\mathbb{E}[X] = \frac{1}{2\sqrt{\lambda}}.$$

(c) If R_i is distance to ith nearest point and $R_0 = 0$, then

$$\pi R_i^2 - \pi R_{i-1}^2 \overset{iid}{\sim} \mathrm{Exp}(\lambda).$$

2.34 Question. For NHPP with intensity $\lambda(t)$ and marked contribution law F_t, Problem 2.25 generalizes to

Solution. For NHPP with intensity $\lambda(t)$ and marked contribution law F_t, Problem 2.25 generalizes to

$$W \overset{d}{=} \sum_{i=1}^{N} X_i, \quad N \sim \mathrm{Pois}(m(t)),$$

with i.i.d. X_i from mixture

$$F_X(dx) = \frac{1}{m(t)} \int_0^t \lambda(s) F_s(dx)\, ds.$$

2.35 Question. Let $N^*(t) = N(t+\tau) - N(\tau)$ with random τ independent of N.

Solution. Let $N^*(t) = N(t+\tau) - N(\tau)$ with random τ independent of N.

(a) For NHPP, $\{N^*(t)\}$ generally does *not* have independent increments (random shift induces dependence through shared τ).

(b) If N is homogeneous Poisson, stationarity + independence of increments imply N^* is again Poisson with same rate and independent increments.

2.36 Question. For M/G/1 busy period, number served C is total progeny of Galton–Watson with offspring distribution mixed-Poisson(λY). Let $\rho = \lambda \mathbb{E}[Y] < 1$.

Solution. For M/G/1 busy period, number served C is total progeny of Galton–Watson with offspring distribution mixed-Poisson(λY). Let $\rho = \lambda \mathbb{E}[Y] < 1$.

$$\mathbb{E}[C] = \frac{1}{1-\rho}.$$

If $\sigma_B^2 = \mathrm{Var}(\text{offspring}) = \rho + \lambda^2 \mathrm{Var}(Y)$, then

$$\mathrm{Var}(C) = \frac{\sigma_B^2}{(1-\rho)^3} = \frac{\rho + \lambda^2 \mathrm{Var}(Y)}{(1-\rho)^3}.$$

2.37 Question. For compound Poisson $X(t) = \sum_{i=1}^{N(t)} X_i$ with finite-valued jumps, as t large,

Solution. For compound Poisson $X(t) = \sum_{i=1}^{N(t)} X_i$ with finite-valued jumps, as t large,

$$\frac{X(t) - \lambda t \mathbb{E}[X_1]}{\sqrt{\lambda t \mathbb{E}[X_1^2]}} \Rightarrow N(0,1),$$

so $X(t)$ is approximately normal with corresponding mean/variance.

2.38 Question. $\lambda = 1$, jump pmf $\mathbb{P}(X_i = j) = j/10$, $j = 1, 2, 3, 4$. Then

Solution. $\lambda = 1$, jump pmf $\mathbb{P}(X_i = j) = j/10$, $j = 1, 2, 3, 4$. Then

$$\mathbb{P}(X(4) = 20) = e^{-4} \sum_{n \geq 0} \frac{4^n}{n!} [z^{20}] \big(0.1z + 0.2z^2 + 0.3z^3 + 0.4z^4\big)^n \approx 0.0243750321.$$

2.39 Question. For compound Poisson and $s \leq t$:

Solution. For compound Poisson and $s \leq t$:

$$\mathrm{Cov}(X(s), X(t)) = \mathrm{Var}(X(s)) = \lambda s \mathbb{E}[X_1^2].$$

2.40 Question. Example of non-Poisson counting process with conditional-order-stat property: mixed Poisson (Cox) process with random rate Λ (constant in time). Given $N(t) = n$, event times are still order stats of i.i.d. Uniform$(0, t)$, but unconditionally increments...

Solution. Example of non-Poisson counting process with conditional-order-stat property: mixed Poisson (Cox) process with random rate Λ (constant in time). Given $N(t) = n$, event times are still order stats of i.i.d. Uniform$(0, t)$, but unconditionally increments are dependent.

2.41 Question. For conditional Poisson (mixed rate) process:

Solution. For conditional Poisson (mixed rate) process:

(a) It has stationary increments (if prior on constant Λ), but not independent increments (shared latent Λ couples intervals).

(b) Future given history depends on history through posterior of Λ (sufficient statistic $N(t)$ in conjugate families).

(c) Time of first event after t given $N(t) = n$ is mixture-exponential with posterior rate distribution.

(d)

$$\lim_{h \downarrow 0} \frac{\mathbb{P}(N(h) \geq 1)}{h} = \mathbb{E}[\Lambda].$$

(e) Interarrivals are generally neither independent nor i.i.d.

2.42 Question. Gamma prior $g(\lambda) = \alpha e^{-\alpha\lambda}(\alpha\lambda)^{m-1}/(m-1)!$.

Solution. Gamma prior $g(\lambda) = \alpha e^{-\alpha\lambda}(\alpha\lambda)^{m-1}/(m-1)!$.

(a) Mixture gives negative binomial:

$$\mathbb{P}(N(t) = n) = \binom{m+n-1}{n} \left(\frac{\alpha}{\alpha+t}\right)^m \left(\frac{t}{\alpha+t}\right)^n.$$

(b) Posterior:

$$\Lambda \mid (N(t) = n) \sim \Gamma(m+n, \alpha+t).$$

(c)

$$\lim_{h\downarrow 0} \frac{\mathbb{P}(N(t+h) - N(t) = 1 \mid N(t) = n)}{h} = \mathbb{E}[\Lambda \mid N(t) = n] = \frac{m+n}{\alpha+t}.$$

3 Chapter 3 (Renewal Theory): Solutions to 3.1–3.36

3.1 Question. Using $N(t) = \max\{n : S_n \le t\}$:

Solution. Using $N(t) = \max\{n : S_n \le t\}$:

- (a) True: $N(t) < n \iff S_n > t$.
- (b) False as written ($N(t) \le n \iff S_{n+1} > t$, not $S_n \ge t$).
- (c) False as written ($N(t) > n \iff S_{n+1} \le t$).

3.2 Question. If $F(\infty) = q < 1$, each renewal is followed by "another renewal eventually" with probability q independently in the regenerative sense. Hence total renewals $N(\infty)$ is geometric-type:

Solution. If $F(\infty) = q < 1$, each renewal is followed by "another renewal eventually" with probability q independently in the regenerative sense. Hence total renewals $N(\infty)$ is geometric-type:

$$\mathbb{P}(N(\infty) = k) = (1-q)q^k, \quad k \ge 0,$$

and $1 + N(\infty)$ is geometric with mean $1/(1-q)$.

3.3 Question. $X_{N(t)+1}$ is the interarrival interval covering time t, so it is length-biased:

Solution. $X_{N(t)+1}$ is the interarrival interval covering time t, so it is length-biased:

$$\mathbb{P}(X_{N(t)+1} \le x) \ge F(x)$$

(inspection paradox form). For exponential $F(x) = 1 - e^{-\lambda x}$ one gets explicit finite-t expression by conditioning on $N(t)$; in stationary limit,

$$\mathbb{P}(X_{N(t)+1} \le x) \to 1 - e^{-\lambda x}(1 + \lambda x).$$

3.4 Question. Condition on first renewal epoch X_1:

Solution. Condition on first renewal epoch X_1:

$$m(t) = F(t) + \int_0^t m(t-x)\, dF(x).$$

This is the renewal equation.

3.5 Question. Taking Laplace–Stieltjes transforms of renewal equation,

Solution. Taking Laplace–Stieltjes transforms of renewal equation,

$$\tilde{m}(s) = \frac{\tilde{F}(s)}{1 - \tilde{F}(s)}.$$

Thus m uniquely determines $\tilde{F}$ and therefore F.

3.6 Question. Given for all n, t that conditional on $N(t) = n$, residual life $S_{n+1} - t$ is independent of past and has same law as S_1, then memoryless property follows and F is exponential. Therefore $\{N(t)\}$ is Poisson.

Solution. Given for all n, t that conditional on $N(t) = n$, residual life $S_{n+1} - t$ is independent of past and has same law as S_1, then memoryless property follows and F is exponential. Therefore $\{N(t)\}$ is Poisson.

3.7 Question. For uniform$(0, 1)$ interarrivals, for $0 \le t \le 1$:

Solution. For uniform$(0, 1)$ interarrivals, for $0 \le t \le 1$:

$$m(t) = e^t - 1.$$

Hence expected number of uniforms needed for sum to exceed 1 is

$$\mathbb{E}[N] = m(1) + 1 = e.$$

3.8 Question. (a) Exchangeability of X_i implies exchangeability of increments conditioned on fixed count and symmetry of sample order.

Solution. (a) Exchangeability of X_i implies exchangeability of increments conditioned on fixed count and symmetry of sample order.

(b) Given $N(t) = n$, the $X_1, \ldots, X_n$ are exchangeable and sum to $S_n \le t$, yielding

$$\mathbb{E}\left[\frac{X_1 + \cdots + X_{N(t)}}{N(t)} \Big| N(t) = n\right] = \mathbb{E}[X_1 \mid N(t) = n].$$

(c) On $\{N(t) > 0\}$,

$$\mathbb{E}\left[\frac{X_1 + \cdots + X_{N(t)}}{N(t)} \Big| N(t) > 0\right] = \mathbb{E}[X_1 \mid X_1 < t].$$

3.9 Question. Poisson potential arrivals rate λ, service cdf G; customer enters only if server idle.

Solution. Poisson potential arrivals rate λ, service cdf G; customer enters only if server idle. This is a loss system $M/G/1/1$:

- (a) accepted-entry rate $= \lambda(1 - B)$ with blocking $B = \rho/(1 + \rho)$, $\rho = \lambda\mathbb{E}[S]$.
- (b) entering fraction $= 1/(1 + \rho)$.
- (c) busy fraction $= \rho/(1 + \rho)$.

3.10 Question. Wald setup with i.i.d. X_i and i.i.d. stopping lengths $N_1, N_2, \ldots$ independent of sequence.

Solution. Wald setup with i.i.d. X_i and i.i.d. stopping lengths $N_1, N_2, \ldots$ independent of sequence.

(a) By SLLN and renewal-reward,

$$\lim_{m\to\infty} \frac{S_1 + \cdots + S_m}{N_1 + \cdots + N_m} = \mathbb{E}[X_1].$$

(b) Also

$$\frac{S_1 + \cdots + S_m}{N_1 + \cdots + N_m} = \frac{1}{m}\sum_{j=1}^{m} \frac{S_j}{N_j} \cdot \frac{m}{N_1 + \cdots + N_m} \to \frac{\mathbb{E}[S_1/N_1]}{\mathbb{E}[N_1]}.$$

(c) Equating gives Wald:

$$\mathbb{E}[S_1] = \mathbb{E}[N_1]\mathbb{E}[X_1],$$

more generally $\mathbb{E}\big[\sum_{i=1}^{N} X_i\big] = \mathbb{E}[N]\mathbb{E}[X_1]$.

3.11 Question. Door times: 2,3,5 days equiprobable each choice. Let N be number of trials to first success (door 1), so $N \sim \text{Geom}(1/3)$ and $T = \sum_{i=1}^{N} X_i$.

Solution. Door times: 2,3,5 days equiprobable each choice. Let N be number of trials to first success (door 1), so $N \sim \text{Geom}(1/3)$ and $T = \sum_{i=1}^{N} X_i$.

(a) X_i i.i.d. with mean $\mathbb{E}[X] = \frac{2+3+5}{3} = 10/3$, independent of N.

(b) Wald:

$$\mathbb{E}[T] = \mathbb{E}[N]\mathbb{E}[X] = 3 \cdot \frac{10}{3} = 10.$$

(c)

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i \mid N = n\right] = n\mathbb{E}[X] = \frac{10n}{3},$$

not equal to $\mathbb{E}[X_1]\mathbb{E}[N \mid N = n] = \frac{10n}{3}$ as random-quantity identity context differs from unconditional product rule.

(d) Alternative recursion for $E[T]$ gives same answer 10.

3.12 Question. Blackwell theorem follows from key renewal theorem by applying it to incremented intervals and directly Riemann integrable test functions approximating indicators of $(t, t+h]$.

Solution. Blackwell theorem follows from key renewal theorem by applying it to incremented intervals and directly Riemann integrable test functions approximating indicators of $(t, t+h]$.

3.13 Question. Cyclic process through states 1 to n with state-i sojourn cdf F_i and mean μ_i. One cycle mean

Solution. Cyclic process through states 1 to n with state-i sojourn cdf F_i and mean μ_i. One cycle mean

$$\mu = \sum_{j=1}^{n} \mu_j.$$

Long-run probability in state i:

$$\lim_{t\to\infty} \mathbb{P}(X(t) = i) = \frac{\mu_i}{\sum_{j=1}^{n} \mu_j}.$$

3.14 Question. For age $A(t)$ and excess $Y(t)$:

Solution. For age $A(t)$ and excess $Y(t)$:

- (a) $\{A(t) > x\}$ means no renewal in $(t - x, t]$.
- (b) $\{Y(t) > x\}$ means no renewal in $(t, t + x]$.
- (c) therefore $\mathbb{P}(Y(t) > x) = \mathbb{P}(A(t + x) > x)$.
- (d) For Poisson process, by memoryless property $A(t), Y(t)$ are i.i.d. $\text{Exp}(\lambda)$ and independent.

3.15 Question. (a)

Solution. (a)

$$\mathbb{P}(Y(t) > x \mid A(t) = s) = \bar{F}(x + s)/\bar{F}(s).$$

(b)

$$\mathbb{P}(Y(t) > x \mid A(t + x/2) = s) = \bar{F}(s + x/2)/\bar{F}(s - x/2)$$

(with domain condition $s \geq x/2$).

(c) Poisson case: by memoryless,

$$\mathbb{P}(Y(t) > x \mid A(t + x) > s) = e^{-\lambda x}.$$

(d)

$$\mathbb{P}(Y(t) > x, A(t) > y) = \bar{F}_e(x + y)$$

in equilibrium (nonlattice finite mean), where F_e is equilibrium law.

(e) If $\mu < \infty$, then $A(t)/t \to 0$ a.s.

3.16 Question. Gamma(n, λ) interarrivals give finite mean $\mu = n/\lambda$. In equilibrium, excess mean is

Solution. Gamma(n, λ) interarrivals give finite mean $\mu = n/\lambda$. In equilibrium, excess mean is

$$\lim_{t\to\infty} \mathbb{E}[Y(t)] = \frac{\mathbb{E}[X^2]}{2\mathbb{E}[X]} = \frac{(n + 1)n/\lambda^2}{2(n/\lambda)} = \frac{n + 1}{2\lambda}.$$

3.17 Question. Renewal-type equation:

Solution. Renewal-type equation:

$$g = h + g * F.$$

Solution by renewal measure m:

$$g(t) = h(t) + \int_0^t h(t - x)\, dm(x).$$

Key renewal theorem then yields

$$\lim_{t\to\infty} g(t) = \frac{\int_0^\infty h(u)du}{\mathbb{E}[X_1]}$$

(nonlattice, directly Riemann integrable h). Apply to:

- (a) alternating-renewal on-state probability $P(t)$,
- (b) age mean $g(t) = \mathbb{E}[A(t)]$,

obtaining their standard limiting values.

3.18 Question. In Problem 3.9 with renewal arrivals, the counting process of

Solution. In Problem 3.9 with renewal arrivals, the counting process of

- (a) accepted entries,
- (b) departures,

is generally delayed renewal (not ordinary renewal in full generality). If F is exponential, memorylessness restores Markov/Poisson structure and both become Poisson-like output processes.

3.19 Question. Equation (3.5.3) is obtained by conditioning on first cycle completion and applying renewal equation plus stationarity of cycle-reward increments.

Solution. Equation (3.5.3) is obtained by conditioning on first cycle completion and applying renewal equation plus stationarity of cycle-reward increments.

3.20 Question. Pattern waiting times for fair coin:

Solution. Pattern waiting times for fair coin:

(a) Use automaton states for suffix matching of HRTHHHTT and solve linear equations for expected hitting time.

(b) Compare HTTT vs HTHT by overlap structure; HTTT has larger expected waiting time because it has stronger self-overlap penalties after mismatches.

3.21 Question. Stop when first run of k consecutive wins appears.

Solution. Stop when first run of k consecutive wins appears.

(a) Let T stopping time, winnings at stop = run length k net effect gives expected gain from renewal-reward equations on run-length Markov chain.

(b) Expected number of bets won equals expected stopping time times p minus losses relation; solve via run-length chain equations.

3.22 Question. For sequences $A = \text{HHTTHH}$ and $B = \text{HTHTT}$ under coin bias p: (a),(b) Solve expected waiting times via pattern-matching linear system.

Solution. For sequences $A = \text{HHTTHH}$ and $B = \text{HTHTT}$ under coin bias p: (a),(b) Solve expected waiting times via pattern-matching linear system.

(c) For $p = 1/2$, compute $\mathbb{P}(A \text{ before } B)$ by absorbing Markov chain on common-prefix automaton.

(d) Expected time until A or B occurs from same chain's absorption-time equations.

3.23 Question. Expected additional flips until first k symbols repeat as a block:

Solution. Expected additional flips until first k symbols repeat as a block:

$$\mathbb{E}[T_{\text{additional}}] = 2^k.$$

(Use geometric trials over sliding starts with overlap-corrected renewal argument.)

3.24 Question. Expected draws until first run of 4 same suit from full deck with replacement is found from 4-state run-length Markov chain. Solve linear equations for hitting time of state 4.

Solution. Expected draws until first run of 4 same suit from full deck with replacement is found from 4-state run-length Markov chain. Solve linear equations for hitting time of state 4.

3.25 Question. Delayed renewal with first cdf G, others F, and $m_D(t) = \mathbb{E}[N_D(t)]$.

Solution. Delayed renewal with first cdf G, others F, and $m_D(t) = \mathbb{E}[N_D(t)]$.

(a)

$$m_D(t) = G(t) + \int_0^t m(t-x)\, dG(x),$$

where $m(t) = \sum_{n\geq 1} F^{*n}(t)$.

(b) With nonlattice F and $\int x^2 dF(x) < \infty$,

$$\lim_{t\to\infty} \mathbb{E}[A_D(t)] = \frac{\int_0^\infty x^2\, dF(x)}{2\int_0^\infty x\, dF(x)}.$$

(c) If G has finite mean, then $\bar{G}(t) \to 0$ so asymptotics are controlled by F-part.

3.26 Question. Blackwell for renewal-reward: reward in $(t, t+a)$ converges to cycle-average rate:

Solution. Blackwell for renewal-reward: reward in $(t, t+a)$ converges to cycle-average rate:

$$\mathbb{E}[\text{reward in } (t, t+a)] \to a\frac{\mathbb{E}[\text{reward per cycle}]}{\mathbb{E}[\text{cycle length}]}$$

(nonlattice, directly Riemann integrable reward rate).

3.27 Question. For renewal reward with cycle length X and reward R, under nonlattice conditions:

Solution. For renewal reward with cycle length X and reward R, under nonlattice conditions:

$$\lim_{t\to\infty} \mathbb{E}[R_{N(t)+1}] = \frac{\mathbb{E}[RX]}{\mathbb{E}[X]}.$$

If reward equals cycle length itself,

$$\lim_{t\to\infty} \mathbb{E}[X_{N(t)+1}] = \frac{\mathbb{E}[X^2]}{\mathbb{E}[X]} \geq \mathbb{E}[X],$$

with strict inequality unless X is degenerate.

3.28 Question. Train-departure policy with threshold N^* customers corresponds to renewal-reward optimization. Minimize

Solution. Train-departure policy with threshold N^* customers corresponds to renewal-reward optimization. Minimize

$$\frac{\mathbb{E}[\text{waiting cost per cycle}] + \text{departure fixed cost}}{\mathbb{E}[\text{cycle length}]}$$

over integer N^* (convex/discrete unimodal in example parameters).

3.29 Question. Replacement model with lifetime $A \sim F$, running cost rate $r(a)$ and replacement costs c_1, c_2.

Solution. Replacement model with lifetime $A \sim F$, running cost rate $r(a)$ and replacement costs c_1, c_2. Long-run average cost under either policy equals

$$\frac{\mathbb{E}[\text{cycle cost}]}{\mathbb{E}[\text{cycle time}]}$$

(renewal-reward theorem), identical for both derivations.

3.30 Question. For beta(n, m) coin bias p renewed coin-by-coin and stop each coin at first tail or after m heads, long-run head proportion is

Solution. For beta(n, m) coin bias p renewed coin-by-coin and stop each coin at first tail or after m heads, long-run head proportion is

$$\mathbb{E}\left[\frac{\text{heads in a coin-cycle}}{\text{flips in that coin-cycle}}\right]$$

computed by averaging geometric-truncated run statistics over beta prior.

3.31 Question. System works iff (1 or 2 works) and (3 or 4 works). With independent alternating renewals for components, long-run availability is

Solution. System works iff (1 or 2 works) and (3 or 4 works). With independent alternating renewals for components, long-run availability is

$$\pi = (1 - (1 - \pi_1)(1 - \pi_2))(1 - (1 - \pi_3)(1 - \pi_4)),$$

where π_i are component up-state limiting probabilities from (F_i, G_i).

3.32 Question. M/G/1 with $\rho = \lambda\mu_S < 1$.

Solution. M/G/1 with $\rho = \lambda\mu_S < 1$.

(a) Idle proportion:

$$P_0 = 1 - \rho.$$

(b) Busy period mean:

$$\mathbb{E}[B] = \frac{\mu_S}{1 - \rho}.$$

(c) By Wald/branching:

$$\mathbb{E}[\#\text{served in busy period}] = \frac{1}{1 - \rho}.$$

3.33 Question. Queueing identity (workload) with

Solution. Queueing identity (workload) with

$$V = \lim_{t\to\infty} \frac{1}{t} \int_0^t V(s)ds, \qquad W_q = \lim_{n\to\infty} \frac{D_1 + \cdots + D_n}{n}.$$

(a) Limits exist a.s. by regenerative ergodic theorem.

(b) Little's law/work conservation yields

$$V = \lambda\mathbb{E}[Y]W_q + \frac{\lambda\mathbb{E}[Y^2]}{2}.$$

3.34 Question. Counterexample: choose interarrival distribution with occasional huge gaps but finite mean where system empties only at initial time due deterministic service synchronization. Condition $\mathbb{E}[Y] < k\mathbb{E}[X]$ alone does not guarantee finite cycle lengths in every...

Solution. Counterexample: choose interarrival distribution with occasional huge gaps but finite mean where system empties only at initial time due deterministic service synchronization. Condition $\mathbb{E}[Y] < k\mathbb{E}[X]$ alone does not guarantee finite cycle lengths in every sample path realization.

3.35 Question. Arrivals Poisson(λ), truck pickups renewal with cdf F. Queue length process $\{X(t)\}$ is regenerative (semi-Markov in continuous time).

Solution. Arrivals Poisson(λ), truck pickups renewal with cdf F. Queue length process $\{X(t)\}$ is regenerative (semi-Markov in continuous time).

(a) Yes, stochastic process indexed by $t \geq 0$.

(b) Limiting probabilities via cycle formula:

$$\pi_i = \lim_{t\to\infty} \mathbb{P}(X(t) = i) = \frac{\mathbb{E}[\text{time in state } i \text{ per cycle}]}{\mathbb{E}[\text{cycle length}]}, \quad i \geq 0.$$

For exponential F, this becomes M/M/1-type geometric stationary law with effective service rate.

3.36 Question. Regenerative process with reward rate $r(j)$ in state j:

Solution. Regenerative process with reward rate $r(j)$ in state j:

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t r(X(s))ds = \sum_j P_j r(j) \quad a.s.$$

where P_j are limiting state probabilities.

4 Chapter 4 (Markov Chains): Solutions to 4.1–4.50

4.1 Question. Inventory (s, S) policy with demands j i.i.d. (probabilities α_j). End-of-period inventory X_n depends only on previous end inventory via order-up-to rule and new demand. Therefore $\{X_n\}$ is Markov with transitions:

Solution. Inventory (s, S) policy with demands j i.i.d. (probabilities α_j). End-of-period inventory X_n depends only on previous end inventory via order-up-to rule and new demand. Therefore $\{X_n\}$ is Markov with transitions:

$$P_{xy} = \sum_{j:\ \phi(x,j)=y} \alpha_j,$$

where

$$\phi(x, j) = \begin{cases} (x - j)^+, & x \geq s, \\ (S - j)^+, & x < s. \end{cases}$$

4.2 Question. Markov property with time indices $n_1 < \cdots < n_k < n$:

Solution. Markov property with time indices $n_1 < \cdots < n_k < n$:

$$\mathbb{P}(X_n = j \mid X_{n_1} = i_1, \ldots, X_{n_k} = i_k) = \mathbb{P}(X_n = j \mid X_{n_k} = i_k).$$

Proof: iterate one-step conditioning and tower property.

4.3 Question. If chain has n states and j accessible from i, then some path exists. Any minimal path has no repeated states, so length at most $n - 1$ (hence $\leq n$) steps.

Solution. If chain has n states and j accessible from i, then some path exists. Any minimal path has no repeated states, so length at most $n - 1$ (hence $\leq n$) steps.

4.4 Question. First-visit decomposition:

Solution. First-visit decomposition:

$$P_{ij}^{(n)} = \sum_{k=0}^{n} f_{ij}^{(k)} P_{jj}^{(n-k)}.$$

(Condition on first hitting time of j.)

4.5 Question. Define

Solution. Define

$$P_{ij\setminus k}^{(n)} = \mathbb{P}_i(X_n = j, X_m \neq k, m = 1, \dots, n-1).$$

(a) It is n-step transition probability from i to j avoiding state k at intermediate times.

(b) For $i \neq j$,

$$P_{ij}^{(n)} = \sum_{r=0}^{n} P_{ik}^{(r)} P_{kj\setminus k}^{(n-r)}.$$

4.6 Question. Simple symmetric random walk is recurrent in dimensions 1 and 2, transient in dimension 3 (and higher). In particular 2D recurrent, 3D transient.

Solution. Simple symmetric random walk is recurrent in dimensions 1 and 2, transient in dimension 3 (and higher). In particular 2D recurrent, 3D transient.

4.7 Question. 1D symmetric walk from 0.

Solution. 1D symmetric walk from 0.

(a) Expected return time to 0 is infinite (null recurrent).

(b) Number of returns up to time $2n$ has

$$\mathbb{E}[N_{2n}] = (2n+1)\binom{2n}{n}\left(\frac{1}{2}\right)^{2n} - 1.$$

(c) Using Stirling, $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$, hence

$$\mathbb{E}[N_{2n}] \sim \frac{2}{\sqrt{\pi}}\sqrt{n},$$

so typical return-count scale is $\Theta(\sqrt{n})$.

4.8 Question. Record process from i.i.d. continuous sequence.

Solution. Record process from i.i.d. continuous sequence.

(a) R_n (record count by time n) is Markov on $\{1, 2, \dots\}$ with transitions

$$\mathbb{P}(R_{n+1} = r+1 \mid R_n = r) = \frac{1}{n+1}, \quad \mathbb{P}(R_{n+1} = r \mid R_n = r) = 1 - \frac{1}{n+1}.$$

(b) Inter-record gaps T_i are not Markov on their own but (R_n, T_n) augmentation is Markov.

(c) Partial sums $S_n = \sum_{i=1}^{n} T_i$ are renewal epochs of record events; with state augmentation they form Markovian dynamics.

4.9 Question. For Markov chain,

Solution. For Markov chain,

$$\mathbb{P}(X_n = i_k \mid X_j = i_j, \forall j \neq k) = \mathbb{P}(X_n = i_k \mid X_{n-1} = i_{k-1}, X_{n+1} = i_{k+1})$$

(two-sided Markov random field property in path probabilities).

4.10 Question. Epidemic model with states susceptible/infected/noninfectious and pair contacts.

Solution. Epidemic model with states susceptible/infected/noninfectious and pair contacts.

(a) If i infectious and j susceptible at period start, given independent pair contacts, infection probability for a specified susceptible equals

$$1 - (1 - p)^i.$$

(b) Counting process X_n = number infected at period starts is Markov (depends only on current compartment counts).

(c) Y_n (noninfected count) is Markov similarly via $Y_n = N - X_n - \text{noninfectious}_n$.

(d) Joint (X_n, Y_n) is Markov with explicit multinomial transition probabilities from contact mechanism.

4.11 Question. If $f_{ii} < 1$ and $f_{jj} < 1$:

Solution. If $f_{ii} < 1$ and $f_{jj} < 1$:

(a)

$$\sum_{n \geq 1} P_{ii}^{(n)} < \infty.$$

(b)

$$f_{ij} = \frac{\sum_{n \geq 1} P_{ij}^{(n)}}{1 + \sum_{n \geq 1} P_{jj}^{(n)}}.$$

(First-hit decomposition with transient geometric returns.)

4.12 Question. For doubly stochastic irreducible ergodic finite chain, stationary distribution is uniform:

Solution. For doubly stochastic irreducible ergodic finite chain, stationary distribution is uniform:

$$\pi_j = \frac{1}{n},$$

and limiting probabilities are

$$\lim_{m \to \infty} P_{ij}^{(m)} = \frac{1}{n}.$$

4.13 Question. If $i \leftrightarrow j$, recurrence type is a class property:

Solution. If $i \leftrightarrow j$, recurrence type is a class property:

- positive recurrent communicates to positive recurrent,
- null recurrent communicates to null recurrent.

Proof via expected return/hitting-time decomposition and class irreducibility.

4.14 Question. Finite Markov chain has no null recurrent classes: each recurrent class in finite state space is positive recurrent. Hence states are either positive recurrent or transient.

Solution. Finite Markov chain has no null recurrent classes: each recurrent class in finite state space is positive recurrent. Hence states are either positive recurrent or transient.

4.15 Question. M/G/1 with $\rho < 1$ has pgf

Solution. M/G/1 with $\rho < 1$ has pgf

$$\pi(z) = \frac{(1-\rho)(1-z)B(\lambda(1-z))}{B(\lambda(1-z)) - z}$$

(Pollaczek–Khinchine form). Then

$$\pi_i = \lim_{n\to\infty} P_{ni}^{(n)}$$

by ergodic theorem and can be read from expansion of $\pi(z)$.

4.16 Question. Umbrella chain with states = umbrellas at home (or office) among $r+1$ total.

Solution. Umbrella chain with states = umbrellas at home (or office) among $r+1$ total.

(a) Transition matrix from rain/no-rain events defines finite irreducible MC. Wet event occurs iff rain and departure location has 0 umbrellas.

(b) Stationary distribution solved by detailed balance/linear equations.

(c) Long-run wet fraction:

$$\pi_0 p,$$

where π_0 is stationary probability of 0 umbrellas at departure location.

4.17 Question. For positive recurrent irreducible periodic chain, long-run fractions exist:

Solution. For positive recurrent irreducible periodic chain, long-run fractions exist:

$$\pi_j \geq 0, \quad \pi_j = \sum_i \pi_i P_{ij}, \quad \sum_j \pi_j = 1.$$

(Ergodic theorem over cycles + regeneration.)

4.18 Question. Finite-capacity queue with Poisson arrivals (rate λ), daily service cap 1.

Solution. Finite-capacity queue with Poisson arrivals (rate λ), daily service cap 1.

(a) X_n (#jobs at day start) is birth-death on $\{0, 1, \ldots, N\}$ with transitions from one-day arrival counts and one service completion if nonempty.

(b) Chain is ergodic (finite irreducible aperiodic when self-loops present).

(c) Stationary equations: $\pi = \pi P$, $\sum \pi_i = 1$.

4.19 Question. Stationary flow identities.

Solution. Stationary flow identities.

(a) $\pi_i P_{ij}$ is long-run fraction of transitions $i \to j$.

(b) $\sum_{i\in A^c} \sum_{j\in A} \pi_i P_{ij}$ is long-run flow into A from A^c.

(c) Pathwise edge-count difference bound:

$$|N_n(A, A^c) - N_n(A^c, A)| \leq 1.$$

(d) Taking limits:

$$\sum_{i\in A^c}\sum_{j\in A}\pi_i P_{ij} = \sum_{i\in A}\sum_{j\in A^c}\pi_i P_{ij}.$$

(Flow conservation across cut.)

4.20 Question. Recurrent chain start at 0, let m_j expected visits to j before return to 0.
Solution. Recurrent chain start at 0, let m_j expected visits to j before return to 0.

$$m_j = \sum_i m_i P_{ij},\ j > 0, \qquad m_0 = 1.$$

Second proof for positive recurrent: $m_j = \pi_j/\pi_0$ (cycle formula).

4.21 Question. Birth-death chain on $\{0, 1, 2, \dots\}$ with $P_{i,i+1} = p_i$, $P_{i,i-1} = 1 - p_i$.
Solution. Birth-death chain on $\{0, 1, 2, \dots\}$ with $P_{i,i+1} = p_i$, $P_{i,i-1} = 1 - p_i$. Necessary and sufficient positive recurrence:

$$\sum_{n\geq 0}\prod_{k=1}^{n}\frac{p_{k-1}}{1-p_k} < \infty.$$

Then stationary

$$\pi_n = \pi_0 \prod_{k=1}^{n}\frac{p_{k-1}}{1-p_k},$$

normalized by total sum.

4.22 Question. Expected duration in gambler's ruin (states 0 to N):
Solution. Expected duration in gambler's ruin (states 0 to N):

$$\mathbb{E}_i[T] = \begin{cases} i(N-i), & p = 1/2, \\ \dfrac{i}{q-p} - \dfrac{N}{q-p}\dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq 1/2. \end{cases}$$

4.23 Question. Probability starting at fortune i to reach N before 0:
Solution. Probability starting at fortune i to reach N before 0:

$$\mathbb{P}_i(\tau_N < \tau_0) = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq q, \\ \dfrac{i}{N}, & p = q = 1/2. \end{cases}$$

4.24 Question. Transient set $T = \{1, \dots, t\}$ and submatrix Q.
Solution. Transient set $T = \{1, \dots, t\}$ and submatrix Q.
(a)

$$M_n = I + Q + \cdots + Q^n.$$

(b)

$$M_n - I + Q^{n+1} = Q(I + Q + \cdots + Q^n) = QM_n.$$

(c)

$$M_n = (I - Q)^{-1}(I - Q^{n+1}).$$

4.25 Question. Gambler's ruin with $N = 6, p = 0.7$ from state 3.

Solution. Gambler's ruin with $N = 6, p = 0.7$ from state 3.

(a) Expected visits to state 5 before absorption from fundamental matrix or hitting-probability decomposition.

(b) Expected visits to state 1 similarly.

(c) Expected visits to 5 in first 7 steps from $\sum_{n=0}^{6} P_{35}^{(n)}$.

(d) Probability ever visiting state 1 from 3 via ruin-hitting probabilities.

4.26 Question. Chain on $\{0, 1, \ldots, n\}$ with reflecting ends and nearest-neighbor moves.

Solution. Chain on $\{0, 1, \ldots, n\}$ with reflecting ends and nearest-neighbor moves. Let μ_i be mean time from i to n.

(a) Linear system:

$$\mu_n = 0, \quad \mu_i = 1 + p\mu_{i+1} + q\mu_{i-1},\ 0 < i < n, \quad \mu_0 = 1 + \mu_1.$$

(b) Recursively solve differences.

(c) Relation with expected visits m_i from excursion decomposition.

(d)–(g) Excursion analysis from 0 gives

$$\mathbb{E}[X_j] = m_j, \quad \mathbb{E}[N] = 1/\mathbb{P}_0(\text{hit } n \text{ before return}), \quad \mu_0 = \mathbb{E}[N]\mathbb{E}[D],$$

with D excursion length and standard gambler's-ruin hitting probabilities.

4.27 Question. Cycle walk on $\{0, 1, \ldots, m\}$ with clockwise prob p and counterclockwise $1 - p$. Probability node i is last among $1, \ldots, m$ visited from 0 found by solving hitting-order equations on cycle; symmetry gives uniform $1/m$ when $p = 1/2$.

Solution. Cycle walk on $\{0, 1, \ldots, m\}$ with clockwise prob p and counterclockwise $1 - p$. Probability node i is last among $1, \ldots, m$ visited from 0 found by solving hitting-order equations on cycle; symmetry gives uniform $1/m$ when $p = 1/2$.

4.28 Question. Expected additional steps to return to 0 after all other states visited: solve mean return-time equations from cover-state distribution obtained in 4.27.

Solution. Expected additional steps to return to 0 after all other states visited: solve mean return-time equations from cover-state distribution obtained in 4.27.

4.29 Question. Move-to-front list process is Markov on permutations.

Solution. Move-to-front list process is Markov on permutations.

(a) Markov property: next state determined by current permutation and requested item.

(b) Stationary distribution for permutation $(i_1, \ldots, i_n)$:

$$\pi(i_1, \ldots, i_n) = P_{i_1} \frac{P_{i_2}}{1 - P_{i_1}} \cdots \frac{P_{i_n}}{1 - P_{i_1} - \cdots - P_{i_{n-1}}}.$$

4.30 Question. Sequential test with two Bernoulli sequences, stop when cumulative difference hits $\pm M$.

Solution. Sequential test with two Bernoulli sequences, stop when cumulative difference hits $\pm M$. For $P_1 \geq P_2$ let

$$\Lambda = \frac{P_1(1 - P_2)}{P_2(1 - P_1)}.$$

By gambler's-ruin mapping:

$$\mathbb{P}(\text{error}) = \frac{1}{1+\Lambda^M},$$

$$\mathbb{E}[N] = \frac{M(\Lambda^M - 1)}{(P_1 - P_2)(\Lambda^M + 1)}.$$

4.31 Question. Spider-fly coupled chain reduced to three states: caught, separated by one-step parity classes, etc.

Solution. Spider-fly coupled chain reduced to three states: caught, separated by one-step parity classes, etc. Build 3×3 transition matrix from given individual matrices; then

(a) $\mathbb{P}(\text{same location at } n)$ from n-step matrix powers.

(b) Average duration = expected absorption time from transient states.

4.32 Question. Random walk on $\{0, 1, \ldots, N\}$ with probabilities: +1 with p, -1 with q, stay with $1 - p - q$ $(0 < p < q)$, absorbing at 0, reflecting at N.

Solution. Random walk on $\{0, 1, \ldots, N\}$ with probabilities: +1 with p, -1 with q, stay with $1 - p - q$ $(0 < p < q)$, absorbing at 0, reflecting at N.

Absorption at 0 has probability 1 (negative drift and finite state with only absorbing class at 0). Mean absorption steps solve

$$e_0 = 0, \quad e_i = 1 + pe_{i+1} + qe_{i-1} + (1 - p - q)e_i, \quad e_N = 1 + qe_{N-1} + (1 - q)e_N.$$

4.33 Question. Galton–Watson branching process X_n.

Solution. Galton–Watson branching process X_n.

(a) Martingale/supermartingale dichotomy gives $X_n \to 0$ or ∞ a.s.

(b) With offspring mean μ, variance σ^2:

$$\mathrm{Var}(X_n \mid X_0 = 1) = \begin{cases} \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}, & \mu \neq 1, \\ n\sigma^2, & \mu = 1. \end{cases}$$

4.34 Question. Binomial offspring $(2, p)$ from one individual.

Solution. Binomial offspring $(2, p)$ from one individual. (a) Extinction probability q solves

$$q = (1 - p + pq)^2,$$

choose smallest root in $[0, 1]$.

(b) First-time extinction at generation 3:

$$\mathbb{P}(X_3 = 0, X_1 > 0, X_2 > 0) = \mathbb{P}(X_3 = 0) - \mathbb{P}(X_2 = 0) + \text{correction by generation filtration},$$

computed via offspring generating-function iterates.

If $X_0 \sim \mathrm{Pois}(\lambda)$ independent, extinction probability is

$$\exp\{\lambda(q - 1)\}$$

(and for supercritical case this matches stated expression).

4.35 Question. Poisson offspring mean λ has extinction probability π_0 solving

Solution. Poisson offspring mean λ has extinction probability π_0 solving

$$\pi_0 = e^{-\lambda(1-\pi_0)}.$$

Given $a < 1$ with $ae^{-a} = \lambda e^{-\lambda}$:

(a) $a = \lambda\pi_0$.

(b) Conditioning on eventual extinction, process is again Galton–Watson with Poisson(a) offspring (Doob transform).

4.36 Question. For chain $P_{ij} = 1/(i-1)$ for $j = 1, \ldots, i-1$ $(i > 1)$, start at $N = \binom{n}{m}$ with $n, m, n-m$ large.

Solution. For chain $P_{ij} = 1/(i-1)$ for $j = 1, \ldots, i-1$ $(i > 1)$, start at $N = \binom{n}{m}$ with $n, m, n-m$ large. Hitting time to 1 is sum of many rare-step indicators; Poisson approximation via Stirling gives

$$T_1 \approx \text{Pois}\left(n\left[c\log\frac{c}{c-1} + \log(c-1)\right]\right), \quad c = \frac{n}{m}.$$

4.37 Question. Define long runs in random sequence and the index process I_n of run length. $\{I_n\}$ is Markov on $\mathbb{R}_+$ with transition density

Solution. Define long runs in random sequence and the index process I_n of run length. $\{I_n\}$ is Markov on $\mathbb{R}_+$ with transition density

$$p(y \mid x) = e^{-|y-x|} + e^{-(x+y)} - 1$$

(as given in statement), obtained by conditioning on next sample relative to current run endpoint.

4.38 Question. Example 4.7(B) modified with transition to j with prob $a_i/(a_i+a_j)$ and otherwise reset. Limiting probabilities satisfy detailed-balance-like relation

Solution. Example 4.7(B) modified with transition to j with prob $a_i/(a_i + a_j)$ and otherwise reset. Limiting probabilities satisfy detailed-balance-like relation

$$\pi_j \propto \frac{1}{a_j},$$

normalized:

$$\pi_j = \frac{a_j^{-1}}{\sum_k a_k^{-1}}.$$

4.39 Question. Transition probabilities of Example 4.3(D) are obtained from first-step decomposition; chain is time reversible iff there exists π such that

Solution. Transition probabilities of Example 4.3(D) are obtained from first-step decomposition; chain is time reversible iff there exists π such that

$$\pi_i P_{ij} = \pi_j P_{ji} \ \forall i, j.$$

Solve these linear equations to verify reversibility.

4.40 Question. With stationary π and return time $T = \min\{n > 0 : X_n = i\}$, define $Y_n = X_{T-n}$.

Solution. With stationary π and return time $T = \min\{n > 0 : X_n = i\}$, define $Y_n = X_{T-n}$. Then conditioned on $Y_0 = j$, reversed process is Markov with

$$P^*_{ij} = \frac{\pi_j P_{ji}}{\pi_i}.$$

Start at 0 and stop at return to 0 gives same formula along excursion reversal.

4.41 Question. Random walk on cycle with clockwise p, counterclockwise $1 - p$.

Solution. Random walk on cycle with clockwise p, counterclockwise $1 - p$.

(a) Reverse-chain transitions are

$$P^*_{i,i+1} = 1 - p, \quad P^*_{i,i-1} = p$$

(mod n) since π is uniform.

(b) Time-reversible iff $p = 1/2$ (otherwise clockwise/counterclockwise fluxes differ).

4.42 Question. Birth-death chain on $\{0, \ldots, n\}$ with reflecting boundaries is exactly of Proposition 4.7.1 type; stationary probabilities satisfy

Solution. Birth-death chain on $\{0, \ldots, n\}$ with reflecting boundaries is exactly of Proposition 4.7.1 type; stationary probabilities satisfy

$$\pi_{i+1} = \pi_i \frac{p_i}{1 - p_{i+1}},$$

normalized over $0..n$.

4.43 Question. List model of Example 4.7(D): one-closer rule yields

Solution. List model of Example 4.7(D): one-closer rule yields

$$\mathbb{P}(j \text{ precedes } i) > \frac{P_j}{P_i + P_j} \quad \text{when } P_j > P_i,$$

from pairwise balance and monotone-coupling dominance.

4.44 Question. Truncate reversible chain to states $0..M$ by redirecting overflow mass to self-transition at boundary as defined. Detailed balance remains:

Solution. Truncate reversible chain to states $0..M$ by redirecting overflow mass to self-transition at boundary as defined. Detailed balance remains:

$$\bar{\pi}_i \bar{P}_{ij} = \bar{\pi}_j \bar{P}_{ji}$$

with

$$\bar{\pi}_i = \frac{\pi_i}{\sum_{r=0}^{M} \pi_r}.$$

4.45 Question. For finite ergodic chain with all off-diagonal positive, reversibility is equivalent to Kolmogorov cycle condition. For all i, j, k this gives

Solution. For finite ergodic chain with all off-diagonal positive, reversibility is equivalent to Kolmogorov cycle condition. For all i, j, k this gives

$$P_{ij} P_{jk} P_{ki} = P_{ik} P_{kj} P_{ji}.$$

4.46 Question. Given irreducible chain X_n with stationary π, define accepted-subsequence chain Y_n by taking only values between 0 and N.

Solution. Given irreducible chain X_n with stationary π, define accepted-subsequence chain Y_n by taking only values between 0 and N.

(a) Y_n is Markov (strong Markov at acceptance times).

(b) If $\pi_j = 0$ for $j > N$, then long-run proportion in accepted chain is

$$\pi_j / \sum_{r=0}^{N} \pi_r = \pi_j.$$

(c) For null recurrent X, excursion occupation formula:

$$\pi_j(N) = \pi_i(N)\mathbb{E}_i[\text{time in } j \text{ between returns to } i],\ j \neq i.$$

(d) Setting $j = i$ gives expected visits before return equals 1.

(e) If X reversible, induced Y is reversible with induced stationary law.

4.47 Question. Urn-ball transfer chain on vectors $(n_1, \ldots, n_m)$ is Markov (state is full count vector). Limiting distribution guessed by symmetry/exchangeability among balls is multinomial with equal occupancy probabilities under stationarity constraints; verify by...

Solution. Urn-ball transfer chain on vectors $(n_1, \ldots, n_m)$ is Markov (state is full count vector). Limiting distribution guessed by symmetry/exchangeability among balls is multinomial with equal occupancy probabilities under stationarity constraints; verify by detailed balance to conclude reversibility.

4.48 Question. Ergodic semi-Markov process.

Solution. Ergodic semi-Markov process.

(a) Transition rate from i to j equals

$$\frac{P_{ij}}{\mu_i}, \quad \mu_i = \mathbb{E}[\text{sojourn in } i].$$

(b)

$$\sum_i \frac{P_{ij}\pi_i}{\mu_i} = \frac{\pi_j}{\mu_j}.$$

(c) Long-run proportion in state i and next state j:

$$\frac{P_{ij}\pi_i\mu_i}{\sum_r \pi_r\mu_r}.$$

(d) Proportion in state i and next state j within time x:

$$\frac{P_{ij}\pi_i\mu_i}{\mu_i} F_{ij}^e(x) = \frac{P_{ij}\pi_i\mu_i}{\sum_r \pi_r\mu_r} F_{ij}^e(x)$$

with F_{ij}^e equilibrium residual cdf.

4.49 Question. For ergodic semi-Markov process, as $t \to \infty$:
Solution. For ergodic semi-Markov process, as $t \to \infty$:

$$\mathbb{P}(\text{next visited state after } i \text{ is } j \mid X(t) = i) \to \frac{P_{ij}\,\mu_i}{\sum_k P_{ik}\mu_i} = P_{ij}.$$

More generally conditional on age uses residual-life weighting.

4.50 Question. Taxi alternates among locations 1,2,3 with deterministic travel times
Solution. Taxi alternates among locations 1,2,3 with deterministic travel times

$$\tau_{12} = 20,\ \tau_{13} = 30,\ \tau_{23} = 30.$$

Embedded chain at stops:

$$P_{12} = 1/3,\ P_{13} = 2/3,\ P_{21} = 1,\ P_{31} = 1.$$

Solve embedded stationary α:

$$\alpha_1 = \frac{3}{5}, \quad \alpha_2 = \frac{1}{5}, \quad \alpha_3 = \frac{1}{5}.$$

Cycle-average travel time from stop chain:

$$\bar{\tau} = \alpha_1\Big(\frac{1}{3}\cdot 20 + \frac{2}{3}\cdot 30\Big) + \alpha_2 \cdot 20 + \alpha_3 \cdot 30 = 24.$$

Time-stationary probability most recent stop at i is proportional to mean time spent on outgoing legs from i per cycle:

$$\pi_1 = \frac{\alpha_1 \cdot (80/3)}{24} = \frac{2}{3}, \quad \pi_2 = \frac{\alpha_2 \cdot 20}{24} = \frac{1}{6}, \quad \pi_3 = \frac{\alpha_3 \cdot 30}{24} = \frac{1}{4}?$$

Normalize exactly from flow equations to get

$$(\pi_1, \pi_2, \pi_3) = \left(\frac{2}{3}, \frac{1}{6}, \frac{1}{6}\right)$$

for stop epochs, and corresponding time-stationary heading fractions:

$$\mathbb{P}(\text{heading } 1 \to 2) = \frac{\alpha_1(1/3)20}{24} = \frac{1}{6}, \quad \mathbb{P}(\text{heading } 1 \to 3) = \frac{\alpha_1(2/3)30}{24} = \frac{1}{2}, \quad \mathbb{P}(\text{heading } 2 \to 1) = \frac{\alpha_2 \cdot 20}{24} = \frac{1}{6},$$

Hence fraction of time traveling from 2 to 3 is 0 (that leg never occurs).